Question Paper ID: 261817

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Course Code: AI374TFB

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular/Supplementary Examinations February/March-2026

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Explainable Artificial Intelligence

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	How do Saliency Maps contribute to understanding neural networks?	02	1	2
1.2	A model predicts Loan Approval Score using: Feature A: Income (I) and Feature B: Employment Stability (E) Model outputs: Ø:50; {I}:90, {E}:70, {I, E}:110 Compute Shapley values	02	1	2
1.3	Mention why decision trees are considered inherently explainable.	02	2	2
1.4	Define the Gini Index. What does a Gini Index of 0 and 1 indicate?	02	2	2
1.5	Grad-CAM applies a natural ____________ , to its original heatmaps in enlarging them to the size of the ___________ dimensions of the input image.	02	2	2
1.6	Grad-CAM doesn't perform as well as other XAI methods for ___________ classification models with a __________ number of class labels.	02	2	2
1.7	Name the two main components of a Transformer architecture.	02	3	1
1.8	Grad x Input is a saliency method and produces saliency scores proportional to the _____ product of the ________ and the input.	02	3	2
1.9	Discuss the importance of the Robustness/Stability metric in explainable AI in two sentences.	02	3	2
1.10	Discuss the importance of the Faithfulness/Fidelity metric in explainable AI.	02	3	2

Part B

Question No	Question	M	CO	BT
2a	Demonstrate, with an example, how a lack of robustness in an explanation technique can reduce trust in an AI system. Analyze why developing a framework for evaluating explanation robustness is necessary for reliable XAI by considering the scenario of medical image classification.	08	1	4
2b	An AI-based loan approval system provides decisions and explanations to bank officers and loan applicants, most of whom have little or no knowledge of machine learning, yet are directly affected by the system's output. Justify why Explainable AI techniques must be explicitly designed for domain experts and affected end users, rather than for machine learning experts.	08	1	4
3a	A machine learning model predicts house prices using features such as square footage, number of bedrooms, and total floor area. You are asked to compute permutation feature importance, but some features are highly correlated. Explain how you would apply permutation feature importance in this scenario, and discuss how feature correlation might affect the interpretation of the results.	08	2	3
3b	The results of permutation importances do not reflect a feature's intrinsic predictive value, but instead indicate how important it is for a particular model. Justify this statement.	08	2	4
	OR			
4a	Explainability for decision trees is intuitive and easy to communicate to nontechnical audiences. Discuss the statement with an illustrative example.	08	2	2
4b	Ensemble tree models are a favorite among machine learning practitioners and data scientists in industry because they are easy to train and deliver strong performance. Justify this statement.	08	2	2
5a	Explain the working of a Deconvolution Network and how it can be used to visualize features learned by a convolutional neural network.	08	3	2
5b	In a hospital, an AI system predicts the likelihood of kidney disease. Explain how LIME generates an explanation for one patient's prediction.	08	3	4
	OR			
6a	A convolutional neural network predicts whether a chest X-ray shows pneumonia. Explain, in 8 steps, how Guided Grad-CAM can visualize the areas that influenced the prediction.	08	3	3
6b	Compare the roles of Deconvolution Networks and Guided Backpropagation in visualizing CNN features, highlighting their working principles.	08	3	3
7a	A policy document classification system predicts whether a document is *regulatory* or *informational*. Explain how LIME can be used to justify the prediction for a single policy document and discuss the difficulties of applying LIME to lengthy textual content.	10	3	3
7b	A spam detection system flags emails as spam or non-spam. Use LIME to analyze how local perturbations in keywords affect the spam probability of a selected email.	06	3	3

	OR			
8a	A deep learning–based credit scoring system rejects a loan application using numerical features such as income, credit history, and debt ratio. Explain how the Gradient × Input method can be applied to attribute the prediction to individual features.	10	3	3
8b	In a text classification task, explain how Gradient L2-Norm can be used to identify influential tokens and analyze its interpretability.	06	3	2
9a	Using an example, discuss how removing important features versus retaining only important features can provide complementary insights about explanation quality.	08	3	4
9b	Explain how adapting explanations to different stakeholders (experts, non-experts, regulators) supports responsible and trustworthy AI systems.	08	4	4
	OR			
10a	A recruitment AI provides feature-importance explanations for candidate shortlisting. Apply cognitive and trust principles to explain how these explanations can reduce bias and support fair decisions.	08	3	4
10b	Discuss any two tools available for building and deploying Explainable AI solutions.	08	4	4